

ORIGINAL RESEARCH

Assessing The Performance of Artificial Intelligence Models In Autism Spectrum Disorder: Accuracy and Readability of ChatGPT, Gemini, and Microsoft Copilot

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Abstract

Objective: Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder affecting social communication and involving restricted, repetitive behaviors. With AI tools like ChatGPT-4, Gemini, and Microsoft Copilot becoming increasingly popular information sources for healthcare professionals and families, this study aimed to evaluate and compare their accuracy and readability when responding to ASD-related questions.

Methods: In this cross-sectional study, we presented 88 questions (45 Frequently Asked Questions [FAQs] and 43 guideline-based) to the three AI models. We sourced questions from social media, parent forums, and clinical guidelines. Two blinded child psychiatrists evaluated response accuracy using a four-grade scale, while readability was assessed using four established indices: Flesch-Kincaid Grade Level, Gunning Fog Index, Coleman-Liau Index, and Flesch Reading Ease.

Results: For FAQs, accuracy rates showed significant differences ($p=0.001$): Gemini (100%), ChatGPT-4 (95.6%), and Microsoft Copilot (71.1%). For guideline-based questions, accuracy also varied significantly ($p=0.010$): Gemini (86.0%), ChatGPT-4 (83.7%), and Microsoft Copilot (55.8%). Interestingly, Microsoft Copilot provided the most readable FAQ responses, while Gemini offered the most balanced readability for guideline-based questions.

Conclusion: Our findings show that Gemini and ChatGPT-4 are highly accurate for ASD information, particularly for complex scientific content, while Microsoft Copilot produced more accessible text despite lower accuracy. These results suggest different models may better serve different audiences—healthcare professionals might benefit from Gemini or ChatGPT-4's precision, while general users might prefer Copilot's readability, highlighting opportunities for improving both reliability and accessibility in healthcare communication.

Keywords: Artificial Intelligence, Autism spectrum disorder, ChatGPT-4, Gemini, Microsoft Copilot, Readability

INTRODUCTION

ASD is a neurodevelopmental disorder that causes ongoing challenges with social communication and interaction. People with ASD also show restricted interests and engage in repetitive behaviors. Individuals with autism spectrum disorder generally face challenges in social interaction, using and understanding nonverbal cues, and developing or maintaining relationships (1,2). ASD typically manifests in early childhood and continues to be a lifelong condition for the individual. While the exact causes of ASD have not been identified, research has identified a combination of genetic-environmental factors as causative in the development of the disorder.

A combination of genetic predispositions, such as family history of neurodevelopmental disorders, and environmental influences, including prenatal and perinatal factors, has been hypothesized to increase the risk for ASD (3). The Centers for Disease Control and Prevention's Autism and Developmental Disabilities Monitoring (ADDM) Network has estimated that approximately 1 in 31 children is identified with ASD (4). ASD is about four times more common in males than in females (5). Following diagnosis, the individual's specific needs are assessed, and they are enrolled in educational support programs aimed at improving social communication and interaction. Parents raising children with ASD often experience considerable stress. They

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Citation: Çalışkan Y, Haşimoğlu A. Assessing the performance of artificial intelligence models in autism spectrum disorder: accuracy and readability of ChatGPT, Gemini, and Microsoft Copilot. Psychiatry and Behavioral Sciences. 2026;16(1):46-56. doi:10.5455/PBS.20250722043208.

Received: Jul 24, 2025

Accepted: Dec 25, 2025



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typically go through several phases: first, facing a lack of information, then actively searching for answers, eventually gathering knowledge, and finally sharing what they've learned with others (6). During this process, they look for information from many sources, including healthcare professionals, publications from healthcare institutions, scientific articles, books, conferences, media, and the internet (7). Research has shown that parents of children with ASD tend to rely heavily on informal sources such as social media and the internet, though the quality of this information can vary (8,9).

More recently, new sources of information based on artificial intelligence models have emerged as increasingly accessible tools for health information seeking. Among these, large language models such as ChatGPT-4 (OpenAI), Gemini (Google), and Microsoft Copilot have gained substantial traction globally (10–12). According to the Stanford AI Index Report 2025, artificial intelligence is now deeply integrated into nearly every aspect of daily life, reshaping sectors including education, finance, and healthcare, where algorithm-driven insights guide critical decisions (13). The scale of adoption is remarkable. As of early 2025, ChatGPT alone surpassed 400 million weekly active users, representing a substantial increase from 300 million users in December 2024, while platforms such as Google Gemini and Microsoft Copilot have similarly experienced rapid adoption (14). In healthcare specifically, the integration of AI has accelerated dramatically. A systematic review of over 500 studies on natural language processing systems in healthcare revealed that the most researched applications focus on enhancing medical knowledge (419 studies) and making diagnoses (178 studies), demonstrating substantial interest from both medical professionals seeking clinical decision support and individuals seeking health information (13). These platforms are now accessible to diverse user groups, including healthcare professionals seeking quick reference information, parents and caregivers looking for guidance on diagnosis and intervention, educators developing support strategies, and individuals with ASD themselves. However, despite their widespread adoption, empirical evaluation of the accuracy, readability, and appropriateness of AI-generated health information, particularly in sensitive and complex contexts like ASD, remains critically limited.

Understanding the accuracy and readability of AI-generated ASD information carries important implications for multiple stakeholders. For healthcare professionals, reliable AI tools could serve as efficient reference

sources during clinical consultations, potentially reducing information gathering time while maintaining the quality of care. For parents and caregivers, who often face overwhelming amounts of conflicting information during critical decision-making periods, validated AI sources could provide accessible, evidence-based guidance when professional consultation is not immediately available. For educators and therapists, standardized AI-generated information could support the development of consistent educational materials and intervention strategies. Moreover, as AI platforms become increasingly embedded in daily information-seeking behaviors, establishing benchmarks for content quality becomes essential for protecting vulnerable populations from potentially misleading or harmful information. Despite the clear need for such validation, empirical evaluation of the accuracy, readability, and appropriateness of AI-generated health information in sensitive contexts like ASD remains critically limited. Recent studies have examined the performance of large language models in various medical domains, including hypertension education (15), breast imaging (16), and neurosurgical board preparation (17), highlighting both strengths and weaknesses in factual accuracy and comprehensibility. However, few studies to date have directly evaluated how well AI-generated ASD content aligns with clinical guidelines or meets the needs of diverse user groups.

Despite the widespread adoption of AI chatbots in healthcare information seeking, comparative analyses evaluating multiple platforms simultaneously using standardized medical content remain scarce. While previous studies have assessed individual AI tools in isolation, the relative performance of leading platforms—ChatGPT-4, Google Gemini, and Microsoft Copilot—has not been systematically examined in the context of ASD. Given their distinct underlying architectures and training methodologies, systematic comparison is essential to provide evidence-based guidance for both clinical practice and patient education. This study addresses this gap by conducting the first comprehensive comparative analysis of these three widely used AI platforms in responding to autism-related questions. Specifically, we evaluate the accuracy of responses to frequently asked questions and scientific queries about ASD, assess the readability of AI-generated content using standardized linguistic indices, and compare performance differences across platforms to inform evidence-based recommendations for healthcare providers, patients, and caregivers seeking reliable autism-related information.

METHODS

We used a cross-sectional diagnostic accuracy design. We focused on non-human subjects and, therefore, were exempt from ethical review. A total of 88 questions related to Autism Spectrum Disorder (ASD) were posed to three large language models (LLMs): ChatGPT-4 (OpenAI), Gemini (Google), and Microsoft Copilot (Microsoft). Questions were developed through a dual approach. Forty-five frequently asked questions (FAQs) were compiled from social media platforms, patient forums, and autism-related online communities. The FAQs were categorized into general information, screening and diagnosis (n=25), etiology and research (n=6), educational approach and treatment (n=8), and prognosis and outcome (n=6). A comprehensive list of all FAQs is provided in Supplementary Table 1. Representative examples include questions such as 'What is autism?', 'What are the early signs of autism?' 'Is there a relationship between autism and vaccines?' 'What is ABA therapy?', and 'Can people with autism spectrum disorder live independently?'

An additional 43 questions were generated from evidence-based clinical guidelines, including the DSM-5 and major child and adolescent mental health recommendations (1,18–29). The scientific questions were further categorized into general information, screening and diagnosis (n=13), etiology and research (n=9), educational approach and treatment (n=15), and prognosis and outcome (n=6). All guideline-based questions are detailed in Supplementary Table 2. Representative examples include questions such as 'In what dimensions does DSM-5 address the clinical features of Autism Spectrum Disorder?' 'What are the tests used in ASD screening?' 'Is there a pharmacological agent with sufficient evidence to justify its use for the core symptoms of autism?', and 'What are the most critical factors that determine the prognosis of children diagnosed with ASD?'

We organized all questions under four thematic categories: (1) General Information, Screening, and Diagnosis; (2) Etiology and Investigations; (3) Interventions and Pharmacological Treatments; and (4) Prognosis and Long-term Outcomes.

To ensure content validity and clinical relevance, all questions were reviewed and refined by a panel of experienced child and adolescent psychiatrists. Expert colleagues provided feedback on phrasing, completeness, and clarity, which was incorporated into the final set of items. The study used publicly accessible,

free versions of each AI model available during the data collection period. Specifically, ChatGPT-4 (version GPT-4, accessed via chat.openai.com) was accessed via OpenAI's web interface, Gemini (version 1.5, accessed via gemini.google.com) via Google's platform, and Microsoft Copilot (version integrated with Bing search, accessed via copilot.microsoft.com) through its web interface. All models were queried between November 10 and November 20, 2024. Each question was entered individually and identically into each model without any additional prompting language (e.g., "summarize" or "in simple terms"). The first full response generated by each model was recorded without follow-up clarification or regeneration.

A total of 264 AI-generated responses from ChatGPT-4, Gemini, and Microsoft Copilot were analyzed and scored by two independent child and adolescent psychiatrists specializing in ASD, each with 7 years of specialist experience. To ensure unbiased evaluation, all AI-generated responses were anonymized, and the psychiatrists were blinded to the AI source of each response. Evaluations were conducted independently using a four-tier grading system: Grade 1 for entirely correct answers, Grade 2 for correct answers with insufficient detail, Grade 3 for responses containing both correct and incorrect information, and Grade 4 for entirely incorrect answers. Inter-rater reliability was assessed using Cohen's kappa coefficient, which yielded a value of 0.816 (95% CI: 0.746–0.886), indicating substantial to almost perfect agreement between raters. Discrepancies occurred in approximately 7.6% of the responses, with all disagreements occurring between adjacent rating categories, reflecting nuanced differences in expert interpretation rather than fundamental scoring inconsistencies. In such cases, both raters reviewed the full content of the AI-generated response together, referring to current ASD guidelines and evidence-based clinical references when necessary. Each evaluator explained their rationale based on the predefined rubric, and consensus was achieved through structured discussion. The final rating was assigned based on agreement following joint review and alignment with the scoring framework. This rigorous and systematic approach ensured that the evaluation process accounted for expert judgment while minimizing subjective biases and maintaining methodological reliability.

The readability of each response was evaluated using four established metrics: Flesch Reading Ease (FRE), Flesch-Kincaid Grade Level (FKGL), Gunning Fog (GF) Index, and Coleman-Liau (CL) Index, which were assessed

using the Readable online platform (30). These metrics quantify different aspects of text complexity through mathematical formulas that consider variables such as sentence structure, word complexity, and syllable distribution. The FRE provides a numerical score between 0 and 100, where elevated values signify enhanced readability. In contrast, the FKGL, GF, and CL indices express readability in terms of educational grade levels within the United States school system, where reduced numerical values correspond to improved accessibility. The FKGL calculates grade level requirements through analysis of sentence length and syllabic complexity per word. The GF Index determines reading difficulty by examining the proportion of multisyllabic terms relative to overall sentence construction. The CL Index distinguishes itself by utilizing character counts rather than syllables, evaluating the mean number of letters per hundred words alongside mean sentence length, which proves particularly valuable when analyzing medical terminology (30).

Statistical Analysis

Statistical analyses were performed using the Statistical Package for the Social Sciences version 27 (SPSS IBM Corp., Armonk, NY). FAQs and scientific questions prepared based on current guideline recommendations were evaluated as separate groups. The scores assigned to the answers are presented as percentages. Hypothesis tests were applied to assess significant differences in the accuracy scores for the scientific questions. For readability analysis, the metrics were reported as means and standard deviations. A p-value of less than 0.05 was considered statistically significant.

RESULTS

The statistical analysis of responses to FAQs revealed significant differences among the AI models ($p = 0.001$). Gemini showed perfect accuracy, providing completely accurate and sufficient answers for 100% ($n = 45$) of all FAQs. ChatGPT-4 achieved 95.6% ($n = 43$) Grade 1 responses, with the remaining 4.4% ($n = 2$) classified as Grade 2. Microsoft Copilot showed more variable performance with 71.1% ($n = 32$) Grade 1 responses, 22.2% ($n = 10$) Grade 2 responses, and 6.7% ($n = 3$) Grade 3 responses.

Pairwise comparisons revealed statistically significant differences between ChatGPT-4 and Microsoft Copilot (95.6% vs. 71.1%, $z = 3.111$, $p = 0.002$), as well as between Microsoft Copilot and Gemini (71.1% vs.

100%, $z = -3.898$, $p = 0.002$). No statistically significant difference was found between ChatGPT-4 and Gemini (95.6% vs. 100%, $z = -1.430$, $p = 0.200$).

When analyzing responses by question category, we observed the following patterns: In the General information, Screening and Diagnosis category ($n = 25$), Gemini achieved 100% accuracy, followed by ChatGPT-4 (96.0%) and Microsoft Copilot (64.0%). The differences between ChatGPT-4 and Microsoft Copilot ($p = 0.010$) and between Microsoft Copilot and Gemini ($p = 0.002$) were statistically significant, while the difference between ChatGPT-4 and Gemini was not ($p = 0.200$). In the Etiology and Investigation category ($n = 6$), both ChatGPT-4 and Gemini achieved 100% accuracy, while Microsoft Copilot reached 83.3%. Despite the 16.7% difference, this was not statistically significant ($p = 0.500$ for ChatGPT-4 vs Copilot, $p = 1.000$ for ChatGPT-4 vs Gemini, $p = 0.500$ for Copilot vs Gemini). For the Intervention and Pharmacological Treatment category ($n = 8$), Gemini achieved 100% accuracy, ChatGPT-4 87.5%, and Microsoft Copilot 62.5%. The comparison between Copilot and Gemini showed a difference of 37.5% ($p = 0.050$), while ChatGPT-4 vs Copilot ($p = 0.100$) and ChatGPT-4 vs Gemini ($p = 0.500$) did not reach significance. In the Follow-up and Prognosis category ($n = 6$), all three models showed identical performance with perfect accuracy (100%) with $p = 1.000$ for all comparisons (Table 1).

The statistical analysis of responses to guideline-based questions revealed significant differences among the AI models ($p = 0.010$). Gemini showed the highest accuracy rate, providing completely accurate and sufficient answers (Grade 1) for 86.0% ($n = 37$) of the questions, with the remaining 14.0% ($n = 6$) classified as Grade 2. ChatGPT-4 achieved 83.7% ($n = 36$) Grade 1 responses and 16.3% ($n = 7$) Grade 2 responses. Microsoft Copilot showed more variable performance with 55.8% ($n = 24$) Grade 1 responses, 34.9% ($n = 15$) Grade 2 responses, and 9.3% ($n = 4$) Grade 3 responses.

Pairwise comparisons revealed statistically significant differences between ChatGPT-4 and Microsoft Copilot (83.7% vs. 55.8%, $z = 2.818$, $p = 0.010$), as well as between Microsoft Copilot and Gemini (55.8% vs. 86.0%, $z = -3.087$, $p = 0.010$). No statistically significant difference was found between ChatGPT-4 and Gemini (83.7% vs. 86.0%, $z = -0.301$, $p = 0.200$).

Table 1. Evaluation of AI model responses to Frequently Asked Questions (FAQs) about autism spectrum disorder

Category	ChatGPT-4			Gemini			Microsoft Copilot			Statistical Significance
	Grade 1 n(%)	Grade 2 n(%)	Grade 3 n (%)	Grade 1 n (%)	Grade 2 n (%)	Grade 3 n (%)	Grade 1 n(%)	Grade 2 n (%)	Grade 3 n (%)	p-value
All Questions (n=45)	43 (95.6%)	2 (4.4%)	0 (0.0%)	45 (100.0%)	0 (0.0%)	0 (0.0%)	32 (71.1%)	10 (22.2%)	3 (6.7%)	0.001*
General information, screening, and diagnosis (n=25)	24 (96.0%)	1 (4.0%)	0 (0.0%)	25 (100.0%)	0 (0.0%)	0 (0.0%)	16 (64.0%)	7 (28.0%)	2 (8.0%)	0.001*
Etiology and investigations (n=6)	6 (100.0%)	0 (0.0%)	0 (0.0%)	6 (100.0%)	0 (0.0%)	0 (0.0%)	5 (83.3%)	1 (16.7%)	0 (0.0%)	0.500
Interventions and pharmacological treatments (n=8)	7 (87.5%)	1 (12.5%)	0 (0.0%)	8 (100.0%)	0 (0.0%)	0 (0.0%)	5 (62.5%)	2 (25.0%)	1 (12.5%)	0.500
Prognosis and long-term outcomes (n=6)	6 (100.0%)	0 (0.0%)	0 (0.0%)	6 (100.0%)	0 (0.0%)	0 (0.0%)	6 (100.0%)	0 (0.0%)	0 (0.0%)	1.000

Pairwise comparisons:

- ChatGPT-4 vs. Microsoft Copilot: $z=3.111$, $p=0.002^*$
- ChatGPT-4 vs. Gemini: $z=-1.430$, $p=0.200$
- Microsoft Copilot vs. Gemini: $z=-3.898$, $p=0.002^*$

*: Statistically significant ($p<0.05$)

Grade 1: Completely correct, Grade 2: Mostly correct but insufficient, Grade 3: Partially correct with misleading content

When analyzing responses by question category for the guideline-based questions, the following patterns emerged: In the General Information, Screening, and Diagnosis category ($n = 13$), both ChatGPT-4 and Gemini achieved identical performance (84.6% Grade 1), while Microsoft Copilot reached 53.8%. Despite the difference, it did not reach statistical significance ($p = 0.100$). In the Etiology and Investigations category ($n = 9$), Microsoft Copilot surprisingly outperformed both ChatGPT-4 and Gemini, achieving 100% accuracy compared to 88.9% for both other models, though this difference was not statistically significant ($p = 0.500$). For the Interventions and Pharmacological Treatments category ($n = 15$), both ChatGPT-4 and Gemini achieved 80.0% accuracy, while Microsoft Copilot reached 53.3%, but the differences were not statistically significant ($p = 0.200$). The most striking difference was observed in the Prognosis and Long-term Outcomes category ($n = 6$), where Gemini achieved 100% accuracy, ChatGPT-4 83.3%, while Microsoft Copilot failed to provide any completely correct answers (0.0% Grade 1, 83.3% Grade 2, 16.7% Grade 3). Despite the small sample size, these differences were statistically significant ($p = 0.002$) (Table 2).

The results of the readability indices for ChatGPT-4, Gemini, and Microsoft Copilot across 45 FAQs about ASD: Microsoft Copilot consistently produced the simplest and most accessible texts, achieving significantly lower scores on FKGL, GF, and CLI, and higher scores on FRE

($p < 0.05$). ChatGPT-4 generated the most complex responses, while Gemini offered a moderate balance between technical detail and clarity. The results of the Readability Indices for ChatGPT-4, Gemini, and Microsoft Copilot across 45 FAQs about ASD are shown in Table 3.

The statistical analysis of responses to guideline-based questions revealed no significant differences among the AI models for most readability metrics ($p > 0.05$). Gemini showed slightly better readability scores, with a Flesch Reading Ease (FRE) score of 70.03 ± 1.73 , followed by Microsoft Copilot (69.74 ± 2.21) and ChatGPT-4 (69.58 ± 2.84). For Flesch-Kincaid Grade Level (FKGL), ChatGPT-4 showed marginally better performance with 11.93 ± 1.53 , compared to Gemini (12.17 ± 0.70) and Microsoft Copilot (12.23 ± 0.76).

Significant differences were observed in the Gunning Fog Index (GF) ($p < 0.001$), where Gemini and Microsoft Copilot (both 13.46) performed better than ChatGPT-4 (14.30 ± 1.34). Similarly, for the Coleman-Liau Index (CLI) ($p < 0.01$), Gemini achieved the best score (9.89 ± 0.75), followed by ChatGPT-4 (10.44 ± 1.51) and Microsoft Copilot (10.51 ± 1.11).

Table 2. Grading of responses by ChatGPT-4, Gemini, and Microsoft Copilot to questions about autism spectrum disorder based on guideline recommendations

Category	ChatGPT-4			Gemini			Microsoft Copilot			Statistical Significance
	Grade 1 n (%)	Grade 2 n (%)	Grade 3 n (%)	Grade 1 n (%)	Grade 2 n (%)	Grade 3 n (%)	Grade 1 n (%)	Grade 2 n (%)	Grade 3 n (%)	p-value
All questions (n=43)	36 (83.7%)	7 (16.3%)	0 (0.0%)	37 (86.0%)	6 (14.0%)	0 (0.0%)	24 (55.8%)	15 (34.9%)	4 (9.3%)	0.010*
General information, screening, and diagnosis (n=13)	11 (84.6%)	2 (15.4%)	0 (0.0%)	11 (84.6%)	2 (15.4%)	0 (0.0%)	7 (53.8%)	5 (38.5%)	1 (7.7%)	0.100
Etiology and investigations (n=9)	8 (88.9%)	1 (11.1%)	0 (0.0%)	8 (88.9%)	1 (11.1%)	0 (0.0%)	9 (100.0%)	0 (0.0%)	0 (0.0%)	0.500
Interventions and pharmacological treatments (n=15)	12 (80.0%)	3 (20.0%)	0 (0.0%)	12 (80.0%)	3 (20.0%)	0 (0.0%)	8 (53.3%)	5 (33.3%)	2 (13.3%)	0.200
Prognosis and long-term outcomes (n=6)	5 (83.3%)	1 (16.7%)	0 (0.0%)	6 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	5 (83.3%)	1 (16.7%)	0.002*

Pairwise Comparisons:

- ChatGPT-4 vs. Microsoft Copilot: $z=2.818$, $p=0.010^*$
- ChatGPT-4 vs. Gemini: $z=-0.301$, $p=0.200$
- Microsoft Copilot vs. Gemini: $z=-3.087$, $p=0.010^*$

*: Statistically significant ($p<0.05$)

Grade 1: Completely correct, Grade 2: Mostly correct but insufficient, Grade 3: Partially correct with misleading content

Table 3. Readability indices for ChatGPT-4, Gemini, and Microsoft Copilot across 45 frequently asked questions about autism

Readability Index	ChatGPT-4 Mean $\pm$ SD	Gemini Mean $\pm$ SD	Microsoft Copilot Mean $\pm$ SD	p-value
Flesch Reading Ease (FRE)	18.36 $\pm$ 9.94	22.27 $\pm$ 12.82	38.12 $\pm$ 11.40	<0.001*
Flesch-Kincaid Grade Level (FKGL)	15.25 $\pm$ 1.95	15.09 $\pm$ 2.58	11.51 $\pm$ 2.26	<0.001*
Gunning Fog Index (GF)	20.08 $\pm$ 2.05	20.00 $\pm$ 2.90	16.16 $\pm$ 2.49	<0.001*
Coleman-Liau Index (CLI)	17.88 $\pm$ 1.68	16.98 $\pm$ 2.20	14.79 $\pm$ 1.90	<0.001*

Pairwise comparisons:

Comparison	FRE	FKGL	GF	CLI
ChatGPT-4 vs. Gemini	$p=0.107$ $d=0.34$	$p=0.752$ $d=0.07$	$p=0.879$ $d=0.03$	$p=0.032^*$ $d=0.46$
ChatGPT-4 vs. Microsoft Copilot	$p<0.001^*$ $d=1.85$	$p<0.001^*$ $d=1.77$	$p<0.001^*$ $d=1.72$	$p<0.001^*$ $d=1.72$
Gemini vs. Microsoft Copilot	$p<0.001^*$ $d=1.31$	$p<0.001^*$ $d=1.47$	$p<0.001^*$ $d=1.42$	$p<0.001^*$ $d=1.06$

*: Statistically significant ($p<0.05$)d: Effect sizes were interpreted according to Cohen's guidelines: small effect ($d = 0.2-0.5$), medium effect ($d = 0.5-0.8$), and large effect ($d \geq 0.8$)

Pairwise comparisons revealed significant differences between ChatGPT-4 and both other models for the Gunning Fog Index ($p < 0.05$), and between Gemini and Microsoft Copilot for the Coleman-Liau Index ($p < 0.05$). Overall, Gemini showed the most balanced readability

profile across all metrics, while ChatGPT-4 and Microsoft Copilot showed more variable performance. The results of the readability indices for ChatGPT-4, Gemini, and Microsoft Copilot across 43 scientific questions about autism are shown in Table 4.

Table 4. Readability indices for ChatGPT-4, Gemini, and Microsoft Copilot across 43 guideline-based questions about autism

Readability Index	ChatGPT-4 Mean $\pm$ SD	Gemini Mean $\pm$ SD	Microsoft Copilot Mean $\pm$ SD	p-value
Flesch Reading Ease (FRE)	69.58 $\pm$ 2.84	70.03 $\pm$ 1.73	69.74 $\pm$ 2.21	>0.05
Flesch-Kincaid Grade Level (FKGL)	11.93 $\pm$ 1.53	12.17 $\pm$ 0.70	12.23 $\pm$ 0.76	>0.05
Gunning Fog Index (GF)	14.30 $\pm$ 1.34	13.46 $\pm$ 0.60	13.46 $\pm$ 1.08	<0.001*
Coleman-Liau Index (CLI)	10.44 $\pm$ 1.51	9.89 $\pm$ 0.75	10.51 $\pm$ 1.11	<0.01*

Pairwise comparisons:

Comparison	FRE	FKGL	GF	CLI
ChatGPT-4 vs. Gemini	$p > 0.05$ $d = 0.19$	$p > 0.05$ $d = 0.20$	$p < 0.05^*$ $d = 0.79$	$p > 0.05$ $d = 0.45$
ChatGPT-4 vs. Microsoft Copilot	$p > 0.05$ $d = 0.06$	$p > 0.05$ $d = 0.24$	$p < 0.05^*$ $d = 0.69$	$p > 0.05$ $d = 0.05$
Gemini vs. Microsoft Copilot	$p > 0.05$ $d = 0.14$	$p > 0.05$ $d = 0.08$	$p > 0.05$ $d = 0.00$	$p < 0.05^*$ $d = 0.67$

*: Statistically significant ($p < 0.05$)

d: Effect sizes were interpreted according to Cohen's guidelines: small effect ($d = 0.2-0.5$), medium effect ($d = 0.5-0.8$), and large effect ($d \geq 0.8$)

DISCUSSION

In this study, we evaluated the responses of ChatGPT-4, Gemini, and Microsoft Copilot to questions about ASD in terms of accuracy and readability. Our results show that these AI models have different strengths: Gemini and ChatGPT-4 performed better in terms of accuracy, while Microsoft Copilot excelled in readability. Particularly in guideline-based scientific questions, Gemini and ChatGPT-4's production of more accurate responses compared to Microsoft Copilot highlights the differences in these models' capacity to process complex health information. In the readability assessment, Microsoft Copilot's production of more accessible content emphasizes the importance of AI selection for the needs of different user groups. These findings have practical significance, as individuals with autism and their families often encounter challenges in accessing timely specialist consultation, leading many to seek information through digital platforms. This study contributes to the growing body of literature by providing a systematic comparison of three widely used AI platforms in the context of autism spectrum

disorder, offering insights that may inform platform selection for both clinical and educational purposes.

In the existing literature, studies on the use of artificial intelligence models in healthcare have increased since 2023, but they remain limited in number. For instance, in a study on endometriosis, ChatGPT-4 demonstrated an accuracy rate of 91.4% in responding to FAQs, while its accuracy dropped to 67.5% for scientifically oriented questions based on the guidelines of the European Society of Human Reproduction and Embryology, indicating challenges in addressing complex, guideline-based inquiries (31). Similarly, Caglar et al. (2023) reported that ChatGPT-4 reached an accuracy rate of 93.6% when answering questions based on the 2022 Pediatric Urology Guidelines of the European Urology Association (32). The pattern of higher accuracy for FAQs compared to guideline-based questions observed in our study aligns with existing literature on AI performance in healthcare domains. Several factors contribute to this performance gap with complex scientific content. Despite extensive training datasets, these models lack access to all available information, particularly paid publications and the most recent

research updates, which may hinder their ability to provide accurate responses. Moreover, the inherent complexity of academic language and the nuanced nature of scientific inquiries further challenge these models' capabilities in delivering precise answers.

However, the strong performance of Gemini and ChatGPT-4 in our study demonstrates that AI models can achieve remarkable accuracy when applied to well-defined, structured domains. This finding is supported by recent research showing AI's effectiveness in specific healthcare applications. Minnis et al. (2024) demonstrated how advanced AI systems could accurately process behavioral information in children, achieving high accuracy in identifying attachment style (33). Similarly, Mukherjee et al. (2023) used BERT and ChatGPT models to analyze parent dialogues for autism symptom detection, achieving 83% accuracy (34). Beyond individual applications, Stade et al. (2024) provided a comprehensive framework for integrating large language models into behavioral healthcare, emphasizing that clinical psychology represents an uncommonly high-stakes application domain for AI systems where responsible development requires nuanced clinical expertise (35). Their staged integration model, progressing from assistive to collaborative and potentially autonomous AI applications, underscores the importance of rigorous evaluation at each level before advancing to greater autonomy. This framework contextualizes our findings, suggesting that while Gemini and ChatGPT-4 demonstrate strong accuracy for ASD-related information, their integration into clinical practice should follow carefully structured developmental stages with appropriate safeguards and oversight. Recent empirical evidence from Heinz et al. (2025) further substantiates Gen-AI's therapeutic potential in behavioral healthcare. Their randomized controlled trial of a Gen-AI-powered therapeutic chatbot demonstrated significant treatment effects for depression, anxiety, and eating disorder symptoms, with large effect sizes (Cohen's $d = 0.627\text{--}0.903$) and therapeutic alliance scores comparable to human-delivered psychotherapy (36). While Heinz et al. evaluated active treatment delivery, our assessment of information accuracy and readability for autism-related content represents a complementary and foundational step toward responsible AI integration in neurodevelopmental healthcare contexts. These studies, combined with our findings, suggest that while AI faces challenges with highly specialized scientific content, it may serve as a reliable information source

for general ASD-related inquiries. This capability is particularly valuable given that parents and caregivers often turn to various sources—including healthcare professionals, scientific articles, and increasingly, the internet—to address information needs about ASD (6–9). While internet sources can vary significantly in accuracy and reliability, AI models offer the advantage of synthesizing information from numerous scientific articles, guidelines, and authoritative websites, with the added capability of providing source citations upon request.

Readability plays a pivotal role in the effective dissemination of healthcare information, ensuring that materials are accessible to a broad audience. Despite its importance, readability studies specifically focusing on AI-generated healthcare responses are still limited. In line with this, the National Institutes of Health (NIH) recommends that patient education materials be written at a sixth- to eighth-grade reading level to ensure clarity for individuals with varying literacy skills (37). Microsoft Copilot, in our study, emerged as the most readable model, producing responses with lower FKGL scores and higher FRE scores, making it accessible to a wider audience. In contrast, ChatGPT-4 generated the most complex responses, requiring higher literacy levels, while Gemini's readability was moderate, balancing complexity and accessibility. These findings align with previous research comparing AI models in healthcare contexts. For example, a study on breast imaging FAQs showed that although ChatGPT-4 performed better in terms of accuracy, Gemini and Copilot generated more readable responses, making them more suitable for a general audience in our study (16). Similarly, in a study on the readability of AI responses for Retinopathy of Prematurity, Gemini and Microsoft Copilot were found to be more readable than ChatGPT-4 (38). This suggests that different AI models offer complementary strengths, with some prioritizing readability over accuracy and others focusing on precise, evidence-based information. While Microsoft Copilot meets NIH's recommended readability standards, it needs to improve its reliability, particularly in scientific contexts. ChatGPT-4 and Gemini excel in producing accurate, evidence-based content, though their use of complex language structures may limit accessibility for some users. Balancing accuracy and readability is therefore crucial in healthcare communication (39), and further developments in AI tools could enable them to better address the diverse needs of healthcare audiences.

The findings of this study have practical implications

for diverse stakeholders in healthcare. Healthcare providers can use these insights to recommend reliable AI platforms when patients seek autism-related information. Patients and caregivers benefit from understanding which platforms offer superior accuracy for their specific information needs, enabling informed decisions when consulting AI tools. The comparative performance metrics can guide AI developers in improving medical content generation, particularly in specialized psychiatric domains where accuracy is critical. Medical educators may integrate these findings into digital health literacy curricula to train future professionals in evaluating AI-generated medical information. Finally, as AI becomes increasingly integrated into health information ecosystems, policymakers can draw upon this evidence to develop appropriate standards for AI-generated medical content in sensitive areas such as mental health.

Although this study is the first to analyze the knowledge and readability levels of ChatGPT-4, Gemini, and Microsoft Copilot regarding ASD, it has certain limitations. First, this study evaluated specific versions of AI models available in November 2024 (ChatGPT-4 version GPT-4, Gemini version 1.5, and Microsoft Copilot with Bing integration). Since the completion of data collection, newer versions have been released, including GPT-4.5 Turbo and Gemini 2.0, which may demonstrate different performance characteristics. The rapid evolution of large language models means that findings from this study represent a snapshot in time rather than a definitive assessment of these platforms' capabilities. Future research should include longitudinal evaluations to track how model updates affect accuracy and readability over time. Second, the exclusive use of English as the evaluation language limits the applicability of the findings to non-English-speaking populations. Additionally, the study examines only three AI models, excluding other emerging tools, which restricts the scope for a more comprehensive comparison. While the questions used in the study evaluate many aspects of ASD, they may not fully reflect the diverse inquiries users might have about the disorder. Furthermore, the study was conducted at a single center, with evaluations performed by only two experts. Although both evaluators are qualified, this approach may introduce bias and impact the robustness of the findings. Future research should consider multi-center studies involving broader populations and a wider range of questions to enhance generalizability. This would reduce bias and provide

more reliable data on the reliability of AI models in delivering ASD-related knowledge.

Despite these limitations, this study provides a comprehensive evaluation of ChatGPT-4, Gemini, and Microsoft Copilot in the context of ASD, filling a significant gap in the literature. ChatGPT-4, Gemini, and Microsoft Copilot have delivered highly accurate responses to frequently asked questions about ASD. Particularly, ChatGPT-4 and Gemini have shown high accuracy in addressing scientific questions, demonstrating promising potential for usability in the healthcare field. By focusing on both accuracy and readability, this study offers valuable insights into the performance of these AI models in disseminating ASD-related information. The study evaluates responses to both frequently asked questions and scientific content, providing data on the models' ability to address general and specialized inquiries. The involvement of experienced child and adolescent psychiatrists specializing in ASD, along with anonymized responses and independent evaluations, further strengthens the findings. This research represents one of the first detailed comparisons of multiple AI models in the context of ASD, making a unique contribution to the field. To the best of our knowledge, no prior studies in the literature have evaluated AI tools specifically for ASD. By examining the accuracy and reliability of responses, this study highlights the potential of AI to improve access to healthcare and deliver reliable medical information.

This study highlights the evolving potential of AI models such as ChatGPT-4, Gemini, and Microsoft Copilot in providing accurate and accessible information about ASD. The findings demonstrate that ChatGPT-4 and Gemini excel in delivering highly accurate and evidence-based responses, while Microsoft Copilot stands out in terms of readability, making it more suitable for general audiences. The balance between accuracy and user accessibility is identified as a critical area for development to meet the diverse demands of healthcare communication. This study fills a significant gap in the literature while highlighting the potential of artificial intelligence tools to facilitate access to information in the healthcare field. Advancements in AI solutions focusing on adaptability and inclusivity offer valuable opportunities to improve the dissemination of health information and provide more effective support to individuals living with ASD and their families.

Funding: The authors received no financial support for the research.

Conflicts of interest: The authors declare that they have no conflicts of interest.

Ethics Committee Approval: This study did not involve the direct collection of primary data from human participants or animals. No patient-identifiable information was used in this study. Therefore, ethical approval from an Institutional Review Board (IRB) or Research Ethics Committee (REC) was not required. However, all analyses and interpretations adhered to the ethical standards set by the 1964 Helsinki Declaration and its subsequent amendments. No informed consent was required as no human participants were directly involved.

Declaration of AI and AI-assisted Technologies in the Writing Process: During the preparation of this manuscript, the authors used Claude (Anthropic) to refine the language. Following the use of these tools, the authors carefully reviewed and edited the content as necessary and take full responsibility for the accuracy and integrity of the final manuscript.

Peer-review: Externally peer-reviewed.

Author Contributions:

Research idea: YC

Design of the study: YC, AH

Acquisition of data for the study: YC, AH

Analysis of data for the study: YC, AH

Interpretation of data for the study: YC, AH

Drafting the manuscript: YC, AH

Revising it critically for important intellectual content: YC, AH

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